Requirements Engineering Processes – 2
The Role Action Perspective

Handout # 6

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Recap of Last Lecture - 1

- We introduced the concept of requirements engineering process
- We discussed inputs and outputs of the requirements engineering process

Recap of Last Lecture - 2

- We introduced high-level activities in the requirements engineering process
  - Requirements elicitation
  - Requirements analysis and negotiation
  - Requirements specification
  - Requirements validation
  - Requirements management

Today's Topics

- Actors and stakeholders in the requirements engineering process
- Process and process improvement for requirements engineering

Who are Actors?

- Actors in a process are the people involved in the execution of that process
- Actors are normally identified by their roles rather than individually, e.g., project manager, purchasing director, and system engineer

Actors in the RE Process - 1

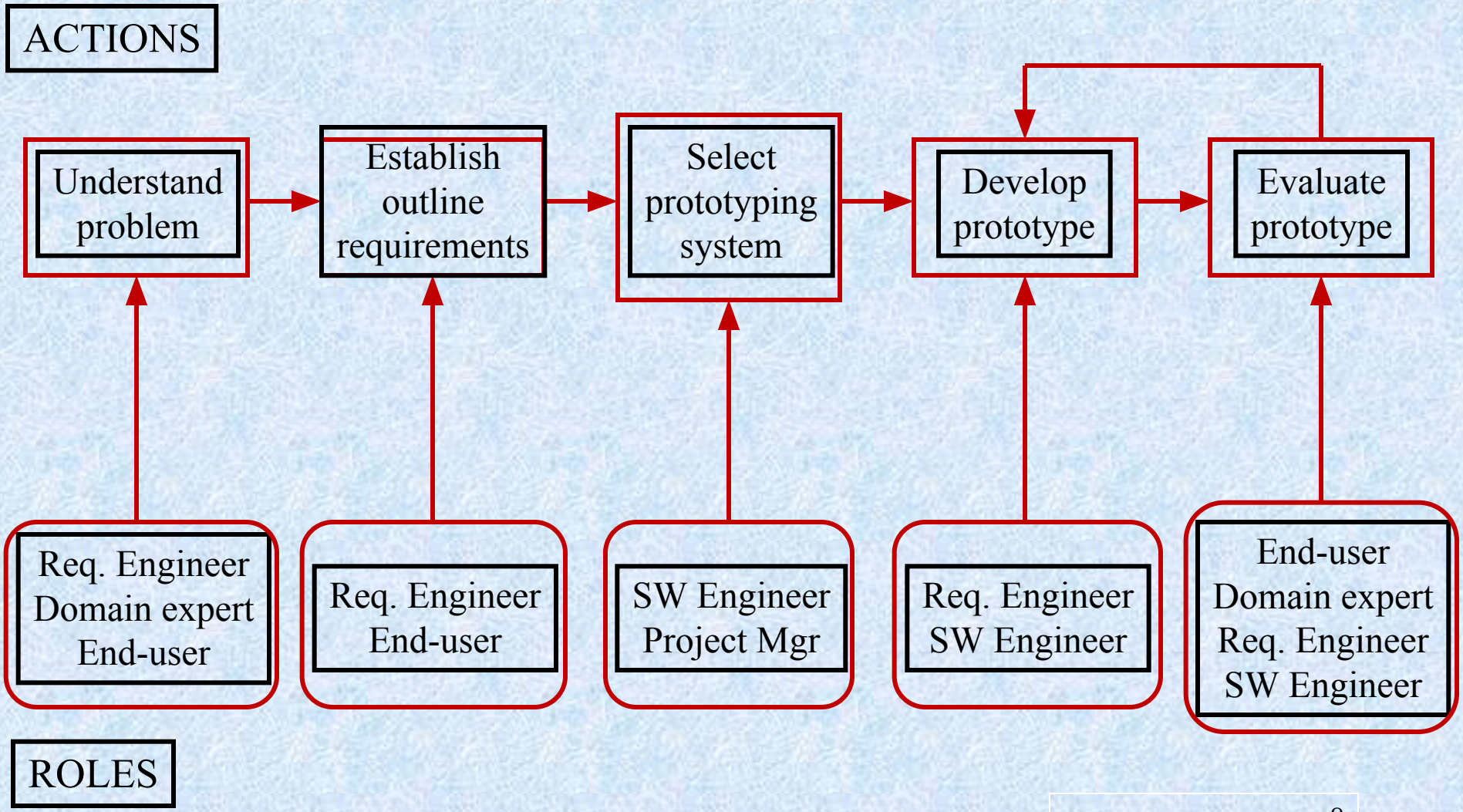
- Requirements engineering involves people
  - who are primarily interested in the problem to be solved (end-users, etc) as well as
  - people interested in the solution (system designers, etc.)

- Another group of people, such as health & safety regulators, and maintenance engineers may be affected by the existence of the system

Actors in the RE Process - 2

- Role-action diagrams are process models which show the actors associated with different process activities
- They document the information needs of different people involved in the process

Role-Action Diagram for Software Prototyping



Role Descriptions - 1

Role	Description
Domain Expert	Responsible for providing information about the application domain and the specific problem in that domain, which is to be solved

Role Descriptions - 2

Role	Description
System End-user	Responsible for using the system after delivery

Role Descriptions - 3

Role	Description
Requirements Engineer	Responsible for eliciting and specifying the system requirements

Role Descriptions - 4

Role	Description
Software Engineer	Responsible for developing the prototype software system

Role Descriptions - 5

Role	Description
Project Manager	Responsible for planning and estimating the prototyping project

Human and Social Factors

- Requirements engineering processes are dominated by human, social and organizational factors because
  - they always involve a range of stakeholders from different backgrounds and with different individual and organizational goals

    e.g. - Doctors doing clinics in the evening

    - Making Rings for bidding
- System stakeholders may come from a range of technical and non-technical background and from different disciplines (e.g. In a hospital)

Stakeholder Types

- Software engineers
- System end-users
- Managers of system end-users
- External regulators
- Domain experts

Factors Influencing Requirements

- Personality and status of stakeholders e.g. Managers
- The personal goals of individuals within an organization
- The degree of political influence of stakeholders within an organization (e.g. Finance department in an organization)

References

- ‘Requirements Engineering: Processes and Techniques’ by G. Kotonya and I. Sommerville, John Wiley & Sons, 1998